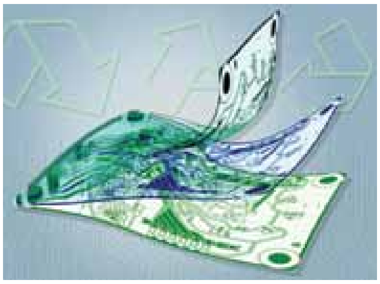
Flexible electronics material reduces e-waste

Researchers at MIT, the University of Utah, and Meta have developed a new flexible substrate material that enhances recyclability and enables scalable manufacturing of complex multilayered circuits. Unlike the commonly used Kapton, which is challenging to reprocess due to its high heat tolerance, the new polyimide-based material cures rapidly at room temperature using ultraviolet light. This innovation simplifies the manufacturing process and allows for the recovery of materials like precious metals and microchips, promoting sustainability in electronics production.



A new kind of flexible substrate material developed at MIT, the University of Utah, and Meta could help combat e-waste (Credit: Christine Daniloff, MIT; iStock)

Fin-like metal nanosheets advance 2D transistors

Researchers at Nankai University have developed a method to integrate ultra-thin insulating layers with two-dimensional (2D) semiconductors, enabling the creation of 2D transistors with electrical capacitance comparable to SiO_2, even below 1-nm thickness. Their approach involves synthesising single-crystalline metal nanosheets, transferred onto 2D substrates, and depositing 2-nm-thick Al_2O_3 or HfO_2 dielectrics. This technique facilitates the development of high-performance top-gated transistors. These transistors exhibit low leakage currents, low operating voltages, and minimal hysteresis. The researchers aim for wafer-scale fabrication of hysteresis-free 2D transistors with sub-nanometre capacitance equivalent thickness.

Bio-inspired vision sensor elevates object detection

Researchers at Hong Kong Polytechnic University have developed a bio-inspired vision sensor capable of detecting objects in diverse lighting conditions, including nighttime, shadows, and fog. This sensor enhances object detection reliability in autonomous vehicles and mobile robots by adapting to the spectral features of its environment. Traditional vision systems struggle under poor lighting, but this new sensor, utilising an array of back-to-back photodiodes with switchable junctions, addresses these issues with low latency and power consumption. Potential applications include autonomous vehicles, medical devices, and surveillance systems. Future research aims to improve the sensor's performance and integrate additional functions.

Solar cell efficiency enhanced by photon splitting

Scientists from UNSW Sydney's School of Chemistry are nearing a breakthrough in next-gen solar cell efficiency by using singlet fission, a process that splits photons into smaller units. The research explores this process to enhance silicon solar cells. Current cells lose much light as heat, capping efficiency at around 27.3%. Singlet fission could boost this by converting high-energy photons more effectively. Funded by ARENA's Ultra Low Cost Solar project, the research aims for over 30% efficiency by 2030. The team used lasers and electromagnets to study singlet fission, aiming to prototype and commercialise improved solar cells.

Enhanced efficiency for solar cells (Credit: https://www.eenewseurope.com)

Micro-sized spectrometer covers visible spectrum

Researchers at the Chinese University of Hong Kong, along with other institutes, have designed and fabricated a new optical spectrometer. This spectrometer uses an organic photodetector with a bias-tunable spectral response, employing a method to manipulate the wavelength-dependent location of photocarrier generation in photodiodes through a trilayer contact consisting of a back contact, an optical spacer, and a back reflector. By integrating this setup with a Schottky diode and an organic bulk heterojunction, the team developed a photodetector. Data from this detector is analysed using a reconstruction algorithm. The spectrometer was tested and found to operate across the visible spectrum (~400–760nm) with resolution.